

# **(TR-104) PYTHON WITH AI/ML**

## **Training Day 76 Report**

**Date: 27 April 2026**

### **Topic Covered: Guardrails Implementation, WebSocket Communication, FastAPI Enhancement & Backend Structuring**

The seventy-sixth day of training focused on implementing AI guardrails, developing real-time WebSocket communication, enhancing FastAPI chat APIs, building alert systems, and improving backend architecture for scalable AI-powered applications.

- Implemented guardrails for input validation, prompt injection detection, query relevance checking, and response safety validation.
- Developed a real-time WebSocket chat system using FastAPI with support for multiple client connections and live response updates.
- Integrated Redis caching to optimize repeated query processing and improve response performance.
- Enhanced REST chat APIs with validation, caching, structured responses, exception handling, and agent integration.
- Built an automated daily alert system for monitoring pending, overdue, and upcoming payments using MongoDB aggregation pipelines.
- Developed a centralized WebSocket connection manager for handling client connections, broadcasting alerts, and managing disconnections.
- Structured FastAPI routes into modular components and implemented separate architectures for REST and WebSocket communication.
- Organized backend modules with centralized logging, reusable utilities, and standardized response formatting.
- Integrated the frontend with backend services to verify real-time alert delivery functionality.
- Reviewed poster content and verified spelling, grammar, and overall presentation quality.
- Revised concepts related to AI guardrails, prompt injection prevention, LangChain agents, WebSocket communication, Redis caching, and FastAPI backend architecture.

Overall, the training enhanced understanding of AI safety mechanisms, real-time communication systems, Redis caching strategies, FastAPI application architecture, and scalable backend development.